Expt No. 07

Sliding Window

Aim:

To write sliding window protocol code using python.

Code: Sender.py

```
import time
import os
ws = int(input("window size"))
msg = input("Enter message:"))
frames=[(i+1, ch) for i, ch in enumerate(msg)]
i=0
while i< len(frames):
    window = frames[i:i+ws]
    with open("Sender_Buffer", "w") as f:
        for no, data in window:
            f.write(f"{no}, {data}\n")
    print("\n Sending :", window)
    time.sleep(2)
    while not os.path.exists("Receiver_Buffer"):
        time.sleep(1)
    with open("Receiver_Buffer") as f:
        ack, no = f.read().strip().split(',');
        no = int(no)
    print("Received", ack, no)
    if ack == "Ack" and no == window[-1][0]:
        i+=ws
    elif ack == "NACK":
        i = no-1
    open("Receiver_Buffer","w").close()
print("\n Transmission Completed")
```

receiver.py

```
import time, os
expected = 1
while True:
    if not os.path.exists("Sender_Buffer"):
        time.sleep(1)
        continue
    with open("Sender_Buffer") as f:
        frames = [(int(x.split(",")[0]), x.split(",")[1].
                   strip()) for x in f if x.strip()]
    if not frames:
        time.sleep(1)
        continue
    print("\n Received", frames)
    for no, data in frames:
        if no != expected:
            with open("Receiver_Buffer", "w") as f:
                f.write(f"NACk, {expected}")
            break
        expected += 1
    else
        with open("Receiver_Buffer", "w") as f:
            f.write(f"ACk, {frame[-1][0]}")
time.sleep(1)
```

Sender Terminal

```
window size: 3    Enter message: Vishnu
Sending [(1,'V'),(2,'i'),(3,'s')]
Received ACk 3
Sending [(4,'h'),(5,'u')]
Received NACK 2
Transmission Completed
```

Receiver Terminal

```
Received: [(1,'V'),(2,'i'),(3,'s')]
Received: [(4,'h'),(5,'u')]
```

Result — Thus the code is implemented for Sliding window.